International University of East Africa, Faculty of Engineering

To: The Dean, Faculty of Engineering
From: Jotham Martin Wambi, lecturer, Machine Learning for Engineers
Courses: CML3111, CML3112 and CML3113 (Year 3)
Date: 13 November 2026
Subject: Final practical project for Semester 1: the questions, the instructions and how it will be taken

This note sets out the final assessment for my three machine learning courses. The project counts for 40 percent of the course mark. There is no written examination. The other 60 percent comes from the five assignments A1 to A5 (20), the five quizzes Q1 to Q5 (20) and Practical Test 1 (20), which have already been marked.

1. When and how it is taken

- **Window.** All three courses sit it online in the same window, from Monday 23 November 2026, 00:00 to Wednesday 25 November 2026, 23:59 (Kampala time). There is no room booking.
- **Time.** Each student works alone for 3 hours, counted on our server from the moment they press Start. The clock cannot be paused. A student who starts after 20:59 on Wednesday only has the time left before 23:59.
- **Setup.** Students use their own laptop and Google account and work in Google Colab. They sign in on the same entry page we used for Practical Test 1, receive their own notebook, and submit it from its last cell. Late submissions are kept but not marked.
- **Support.** I answer questions about wording and technical problems on the class channel from 08:00 to 20:00 on each day of the window. If a student's laptop or connection fails, they report it at once with a screenshot and I can give one fresh attempt, still ending at the window's close.
- **Rehearsal.** Students get an unmarked practice version right after Class 3: CML3112 on Tuesday 17 November 2026, CML3113 and CML3111 on Friday 20 November 2026. It uses the same file with a different target, so it does not give away the real questions.

2. The data

Every student gets the same real dataset: one year (2018) of electricity readings from a steel plant in Gwangyang, South Korea, taken every 15 minutes, 35,040 rows in all. It is published on the UCI Machine Learning Repository under a CC BY 4.0 licence (Sathishkumar V E, Shin and Cho, 2021, doi:10.24432/C52G8C). The file is stored inside each student's notebook and checked when it is written out, so nothing needs to be downloaded during the window.

I chose real data on purpose. Students have to deal with the things real data does: the last reading of each day is stamped with the wrong date, one reading is all zeros because the meter recorded nothing, and one column is calculated from the very quantity they are asked to predict, so using it would be cheating without knowing it. I checked the file myself and nothing has been added to it or changed.

3. The questions

Students act as analysts for the plant's energy manager and answer two questions.

- **Question 1.** How much energy will the plant use in each 15-minute period, in kWh, for months the model has not seen? Students compare three models (a simple baseline that predicts the average, linear regression and a decision tree) and report the errors.

- **Question 2.** Should a reading raise a warning? A reading at or above 60 kWh in a quarter hour is a warning, since that is when peak-demand charges build up. Students estimate a probability for each reading with logistic regression, choose a cut-off using a table of costs (a missed warning costs five times a false alarm), and report how many warnings they catch and miss.

They train on January to August, check on September and October, and keep November and December as a test set that they may open only once, after they have chosen their model. They finish with a short decision memo for the manager. The 60 kWh limit, the costs and the manager's limits on error (10 kWh) and response time (50 ms) are set by me for teaching and are stated as such in the brief.

4. Tasks and marks

Part A is the coded notebook, 30 marks. Students write the code for each step as a function.

Part A task	Marks
Clean the rows (dates, an empty reading, impossible values, a column that leaks the answer)	3
Split the year by month into training, validation and test sets	2
Train a baseline model, linear regression and a decision tree	4
Measure the error: MAE and RMSE	2
Warning probabilities with logistic regression	2
Confusion matrix, precision, recall, F1, cost and AUC at a chosen threshold	4
Cross-validation by month	4
Results on the data file	1
The test-set result for the chosen model	2
Test set opened once, after the choice	2
No data leakage: preprocessing fitted on training rows only	2
Reproducible with a fixed random seed	1
Two labelled charts and the required functions present	1
Total	**30**

Part B is the decision memo, 15 marks, written as a short form inside the notebook.

Part B field	Marks
Recommended model and decision threshold	3
Validation error of the three models	3
Test-set error and warning recall	2
Cross-validation mean error	1
Three true limits of the study, chosen from ten statements	3
Best next step, chosen from five	1
Two short written answers: why this model, and why the test set is opened only once	2
Total	**15**

The 45 marks are scaled to 40 percent. Everything tested was taught in Weeks 1 to 10, up to Class 3. The material from Class 4 (Week 11) is left out, because two of the three classes meet for it after the window.

5. Rules for students

- **Allowed:** Colab, the course slides, their lab notebooks, the revision guide and their own notes.
- **Not allowed:** Any AI tool (ChatGPT, Gemini, Copilot or similar) and any help from another person. Students sign a declaration when they submit that the work is their own.

- **Own cells:** Students may write their answers in cells of their own. A correct answer earns its marks wherever it is in the notebook. I applied the same rule when re-checking Practical Test 1 and the coding assignments.

6. Marking

- **Automatic.** The notebooks are marked by computer, as Practical Test 1 was. Each student's functions are run on rows they have never seen, so they need working code, and copying another student's printed numbers earns almost nothing (1 mark of 30).
- **Tested.** Before release I ran the marking on 149 prepared notebooks, correct, partly correct, wrong and dishonest ones, and each received the mark it should.
- **Honesty.** The computer compares all notebooks for copied code and copied written answers, and checks each notebook against the identity used at entry. I look at every flagged case myself. Because the project is unsupervised, I may also ask any student to explain or change their code in a short online call after the window.
- **Results.** Marks will be final by Wednesday 2 December 2026, after questions and checks are settled. Each student receives written feedback for each task.

7. Still to be done before the window

- Set up the submission forms and run the whole process once in Colab myself, from entry to marked result.
- Confirm the 08:00 to 20:00 support hours with the three classes.
- Any student with approved extra time or another arrangement: I would be grateful to know by 20 November so it can be set up.

Course lecturer	Head of Department	Dean, Faculty of Engineering
Signature and date	Signature and date	Signature and date